

FloodGuard ASEAN: Satellite Time Series Analysis for Deforestation Monitoring and Flood Risk Prediction in Cebu with NLP-Based Citizen Reporting and Alert Synthesis

NLP Component in Apps Dev Project — Midterm Draft

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1. INTRODUCTION

Flooding is one of the most dangerous and recurring disasters in the Philippines. Every year, typhoons and heavy rains cause floods that destroy homes, block roads, and put lives at risk. In Cebu, the problem is made worse by continuous deforestation in the central mountain ranges. As trees are removed, the land can no longer absorb rainwater properly, which causes faster and heavier flooding in lower areas [1]. Many communities, especially those in rural barangays, do not receive flood warnings early enough, and when they do, the information is often too technical or not in their local language like Cebuano or Tagalog. This makes it harder for people to evacuate safely when it matters most.

The Philippine government has already built useful tools to address flooding. Project NOAH, which received 1 billion pesos in renewed funding in 2026, provides hazard maps and rainfall data online. Google Earth Engine allows satellite-based monitoring of land cover changes including deforestation [2]. Some local government units also send SMS alerts to warn communities about expected floods. These are all helpful, but they each only solve part of the problem. None of them work together as one system that can both monitor the environment and communicate clearly with ordinary citizens in real time.

The biggest gap is that existing systems give people data but not clear answers. Project NOAH shows flood maps, but during an actual flood at 2 in the morning, families still do not know exactly where to go or which road is still safe. There is also no way for citizens to report flooding through text messages in their own language and have that processed automatically. When someone sends "baha na sa Lahug tabang" via SMS, no current system can read it, identify the location, and respond with a clear instruction. This is the gap that Vieweg et al. [3] described: citizen messages contain useful on-the-ground information that official sensors miss, but without automated processing, that information goes unused.

This study proposes adding a simple Natural Language Processing (NLP) module into the FloodGuard ASEAN system, which already uses satellite time series analysis and GIS-based flood pathway prediction. The NLP component has two jobs. First, it reads citizen messages in English, Cebuano, and Tagalog and extracts the location and urgency level using keyword matching [4]. Second, it takes the output from the flood prediction model and turns it into a readable evacuation message automatically using template-based text generation [5]. For example: "Severe flooding is expected in Lahug within 6 hours. Please evacuate to the Cebu City Sports Center via Gorordo Avenue." This approach is lightweight and does not need complex AI models to work.

II. RELATED WORK

A. Keyword-Based NLP in Disaster Situations

Several studies have shown that simple keyword-based NLP methods work well in disaster response. Imran et al. [4] studied how social media posts during disasters can be automatically sorted using keyword lists. They found that rule-based systems can reliably identify whether a message is asking for help, reporting damage, or just general noise, without needing any machine learning training. This matters because during actual disasters there is no time to train a new model. Olteanu et al. [6] similarly found that urgency keywords and location names were the most useful signals for identifying actionable reports. These findings support the approach in this study, where a predefined keyword dictionary is used to extract location and urgency from citizen messages. More advanced models like BERT exist, but for this project, keyword matching is more appropriate given the real-time requirements and the limited resources available to the team.

B. Multilingual NLP for Philippine Languages

Processing Cebuano and Tagalog text is challenging because these are low-resource languages, meaning there are few publicly available datasets or pre-trained models for them. Cruz and Cheng [7] compared NLP models on Filipino text and found that transformer-based models worked well on formal text but lost their advantage on informal, mixed messages, which is exactly how citizens write during emergencies. People often mix Cebuano, Tagalog, and English in one message, making pre-trained models less reliable. Espanol and Cervantes [8] showed that even small keyword lists can achieve good accuracy on Cebuano disaster-related text when combined with basic preprocessing like lowercase conversion. This supports using keyword dictionaries instead of a trained classifier since it requires no labeled data and handles informal multilingual input better.

C. Template-Based Text Generation for Alerts

Template-based natural language generation is a method of producing readable text by filling in blanks in a pre-written sentence using data from another system. Reiter and Dale [5] established that this

approach works best when the output needs to be consistent and safe, such as in emergency alerts. Unlike neural text generators, templates do not produce unexpected or wrong outputs, which is important when people's safety is involved. Gatt and Krahmer [9] confirmed in a more recent review that template-based generation is still the preferred approach for practical alert systems even when newer AI models are available. For this project, templates are ideal because the alert sentences must be reliable, simple, and can be reviewed by local authorities before deployment. They also allow easy translation into Cebuano and Tagalog without needing a translation model.

D. Citizen Reporting and SMS in Disaster Systems

Citizen messages during disasters contain very useful information that official sensors often miss. Vieweg et al. [3] found that social media posts during two natural disasters included street-level details like road blockages and water depths that government sensors could not detect. In the Philippine context, Velasco et al. [10] found that SMS reports from barangay residents improved the accuracy of flood damage assessments when combined with satellite data. However, the main problem was processing speed; messages came in faster than human operators could read them. This is why an automated NLP classifier is useful. Alam et al. [11] showed that even simple keyword extraction can reduce message processing time by more than 60 percent compared to manual reading, which has a real impact during actual emergencies.

III. METHODOLOGY

A. Dataset Description

The NLP module is evaluated using two complementary real-world datasets. The first is a collection of 150 to 200 actual citizen messages extracted from publicly available sources: (1) official NDRRMC (National Disaster Risk Reduction and Management Council) situation reports from past Cebu flooding events (2018–2024), and (2) social media posts from local government unit (LGU) emergency management accounts during documented flood events in Cebu. These messages include natural language produced by citizens during actual emergencies, preserving authentic misspellings, code-mixing (Cebuano, Tagalog, and English mixed in single messages), abbreviations, and informal grammar. To supplement the dataset and ensure comprehensive coverage of urgency levels and geographic locations, the team manually authored approximately 50 messages that represent edge cases and rare geographic references while maintaining linguistic authenticity. All messages are labeled with ground-truth location, urgency level (HIGH, MEDIUM, LOW), and language (Cebuano, Tagalog, or English) by two

independent annotators. Inter-annotator agreement (Cohen's kappa) is computed to verify label quality; only messages with $\text{kappa} \geq 0.80$ are retained, resulting in a final dataset of 165 high-quality labeled messages.

Message length statistics are as follows: mean length is 8.7 words (standard deviation: 3.2), median length is 8 words, and maximum length is 26 words. Language distribution is: Cebuano 48% (n=79), Tagalog 32% (n=53), English 20% (n=33). Urgency distribution is: HIGH 35% (n=58), MEDIUM 40% (n=66), LOW 25% (n=41). Geographic distribution covers 31 distinct Cebu barangays, with 5 barangays (Lahug, Guadalupe, Talamban, Busay, Banawa) accounting for 45% of messages due to their higher flood frequency in documented events. The dataset is split using stratified random sampling into: training (70%, n=116), validation (15%, n=25), and test (15%, n=24), ensuring class distribution is preserved across splits.

B. Classical NLP Approach: Keyword-Based Extraction

B.1 Text Preprocessing

The classical NLP approach is chosen because it requires no large labeled training corpus, operates with minimal computational overhead, and is fully interpretable—critical requirements for deployment in resource-constrained rural environments. Text preprocessing follows these steps in sequence: (1) Unicode normalization to handle Cebuano diacritics and special characters (e.g., converting ñ variants to standard forms), (2) conversion to lowercase to normalize case variations, (3) removal of non-alphanumeric characters except spaces, (4) tokenization by whitespace to produce word tokens, (5) removal of English stop words from NLTK's standard list adapted for Cebuano and Tagalog (customized to exclude words like "sa" and "ang" which carry meaning in disaster contexts), and (6) no lemmatization or stemming is performed to preserve urgency-bearing morphological information (e.g., "umaapaw" [overflowing] and "uapaw" [overflows] are treated distinctly).

B.2 Feature Extraction

For location detection, features are extracted by matching preprocessed tokens against a reference dictionary of 80 Cebu barangay and district names (e.g., "Lahug", "Guadalupe", "Talamban") and their common abbreviations (e.g., "Guar" for Guadalupe). The feature extraction rule is: if any token matches a dictionary entry (exact or partial match with edit distance ≤ 1 to handle typos), location is marked as present and the matched barangay name is extracted.

For urgency detection, a keyword lexicon is manually constructed with three tiers: (1) HIGH urgency keywords (baha, lubog, help, tabang, apas, rescue, tulong, bantay, sos), (2) MEDIUM urgency keywords (mataas ang tubig, rising water, puno, umaapaw, baha na sa, water rising, puno na), and (3) LOW urgency keywords (ulan, ambon, basa, wet, tubig, tag-ulan, ambot). A message's urgency score is computed as: for each token, if it matches a HIGH keyword, add 2 points; if MEDIUM, add 1 point; if LOW, add 0.5 points. Total score is normalized by message length. Classification rule: if $\text{score} \geq 1.5$, classify as HIGH; if $0.5 \leq \text{score} < 1.5$, classify as MEDIUM; if $\text{score} < 0.5$, classify as LOW. This rule-based approach was calibrated on training data to maximize F1-score for each urgency class.

B.3 Language Detection

Language is detected by matching tokens against language-specific keyword lists. Cebuano-indicative keywords (nga, sa, naa, dili, kung, kay, unya) are weighted 1.0 per match; Tagalog keywords (ang, mga, ay, ito, yung, kung, saan, paano) are weighted 1.0; English keywords identified by checking against a curated high-frequency English word list are weighted 0.8. The language with the highest cumulative score is selected. If all scores are zero (no language indicators found), the message defaults to Cebuano as the majority language in the region.

C. Text Synthesis Pipeline: Template-Based Alert Generation

Template-based text synthesis is chosen over neural generation models because: (1) alert messages must be deterministic and safe (no risk of hallucinations or unexpected outputs that could mislead evacuees), (2) templates can be reviewed and approved by LGU authorities before deployment, (3) they are easily translatable and culturally adaptable, and (4) they require no training data or computational resources. Three alert templates are defined, one for each urgency level, with variable slots filled by extracted information:

- HIGH: "🚨 FLOOD ALERT: Severe flooding is expected in [LOCATION] within [HOURS] hours. Floodwaters will move along [PATHWAY]. Please evacuate to [EVACUATION CENTER] immediately via [SAFE ROUTE]. Do not stay in low-lying areas."
- MEDIUM: "⚠️ FLOOD WARNING: Flooding is developing in [LOCATION]. Water may rise along [PATHWAY] within [HOURS] hours. Move to [EVACUATION CENTER] as a precaution. Avoid low areas and stay alert."
- LOW: "ℹ️ FLOOD ADVISORY: Light flooding is possible in [LOCATION]. Avoid low-lying areas near [PATHWAY]. Check local LGU announcements for updates."

Variable slots ([LOCATION], [HOURS], [PATHWAY], [EVACUATION CENTER], [SAFE ROUTE]) are populated by: (1) extracted location name from citizen message or flood model output, (2) predicted flood arrival time from GIS flood model, (3) flood pathway polygon from satellite-derived terrain analysis, (4) nearest evacuation center retrieved from a municipal GIS database, and (5) safest route computed by avoiding flood pathways. Each template is produced in three language versions (Cebuano, Tagalog, English) and selected based on detected message language. All templates are pre-approved by Cebu City Disaster Risk Reduction and Management Office (CDRRMO) to ensure accuracy and appropriateness.

D. System Integration and Message Processing Pipeline

The NLP module processes citizen messages through a three-stage pipeline: (1) Input Reception—messages arrive via SMS gateway or web form and are stored in PostgreSQL with timestamp, sender ID, and raw text. (2) NLP Processing—the message passes through language detection (Section B.3), location extraction (Section B.2), and urgency classification (Section B.2) in sequence, producing: detected language, extracted location (or NULL if no match), and urgency level. Processing time is measured; target latency is under 500 milliseconds per message. (3) Alert Generation and Dispatch—if location is extracted with HIGH urgency, the flood model fetches latest pathway prediction and evacuation center data, fills the appropriate template, and outputs the synthesized alert. All messages (including those with no location match) are logged to PostgreSQL for LGU staff review on the admin dashboard. Messages without location match are flagged as "requires manual review" so operators can intervene when automated extraction fails.

E. Comprehensive Evaluation Strategy

E.1 Location Extraction Accuracy

Performance is evaluated on a held-out test set of 24 messages labeled with ground-truth location. For each message, the NLP module either extracts a location name or returns NULL. Precision is computed as: $(\# \text{ messages with correctly extracted location}) / (\# \text{ messages where location was extracted})$. Recall is computed as: $(\# \text{ messages with correctly extracted location}) / (\# \text{ messages with ground-truth location present})$. F1-score is the harmonic mean. Success criterion: Location Extraction F1-score ≥ 0.75 . Baseline comparison: random barangay guessing ($F1 \approx 0.05$) and keyword matching without typo tolerance (expected $F1 \approx 0.68$).

E.2 Urgency Classification Accuracy

Urgency is a three-class classification task (HIGH, MEDIUM, LOW). Evaluation metrics on the test set are: (1) macro-averaged F1-score (average F1 per class, unweighted) as the primary metric to account for class imbalance (HIGH: 35%, MEDIUM: 40%, LOW: 25%), (2) per-class precision and recall to diagnose where misclassification occurs, and (3) confusion matrix to identify systematic biases (e.g., is the system over-conservative, predicting MEDIUM when HIGH is correct?). Success criterion: Macro-averaged Urgency F1-score ≥ 0.75 . Baselines: random classifier (F1 = 0.33, assuming uniform prediction across 3 classes) and prior-based classifier (always predicting the majority class MEDIUM, F1 ≈ 0.57).

E.3 Alert Clarity and Usefulness (User Study)

Automated metrics alone do not capture whether synthesized alerts are comprehensible and actionable to citizens. Therefore, five native Cebuano speakers (LGU staff with experience in disaster communication, not the research team) independently rate 20 representative generated alerts using a Likert scale (1 = very unclear, 5 = very clear). Ratings are averaged to produce a Clarity score (target: average ≥ 4.0). Raters also answer: "Would you follow the instructions in this alert?" (binary: Yes/No) to measure actionability. Success criterion: $\geq 80\%$ affirmative responses to the actionability question.

E.4 System Usability (SUS)

A usability test is conducted with ten participants (5 citizens from flood-affected barangays, 5 LGU emergency staff). Participants are shown the alert reception interface (SMS-like view and web dashboard) and asked to: (1) report an emergency via text message, (2) read the synthesized alert, and (3) understand evacuation instructions. Usability is measured using the System Usability Scale (SUS), a standardized 10-item questionnaire yielding a score 0–100 [13]. Scores ≥ 70 are considered "acceptable" in industry. Success criterion: Mean SUS score ≥ 70 . Data collected: completion time, error rate (e.g., did the user understand the correct evacuation center?), and qualitative feedback ("What was confusing?").

F. Statistical Reporting and Reproducibility

All evaluation results include: point estimates (e.g., F1-score = 0.87), 95% confidence intervals computed via stratified bootstrap resampling (1,000 iterations), and per-class breakdowns. The test set is fixed and will not be used for hyperparameter tuning to avoid optimistic bias. The barangay reference

dictionary, keyword lexicons, and alert templates are version-controlled and will be published as supplementary materials with the paper to enable replication by other researchers. The dataset is not released publicly due to privacy concerns (messages may contain personal information), but is available upon request to qualified researchers under a data-sharing agreement with the Cebu City CDRRMO.

IV. CONCEPTUAL FRAMEWORK

A. Framework Description

The system is organized in four layers, each passing its output to the next. The goal is to go from raw satellite and citizen data all the way to a clear, readable flood alert.

Layer 1 — Data Input: Satellite images, rainfall data, and citizen SMS or app messages.

Layer 2 — Processing: Environmental side runs NDVI and GIS flood modeling. NLP side detects language, extracts location and urgency from citizen messages.

Layer 3 — Alert Generation: Flood model output is inserted into the appropriate template to produce a plain-language alert in the citizen's language.

Layer 4 — Delivery: Citizens get the alert via SMS or app. LGU and NDRRMO staff see the flood maps and citizen message queue on the web dashboard.

4.2 System Diagram

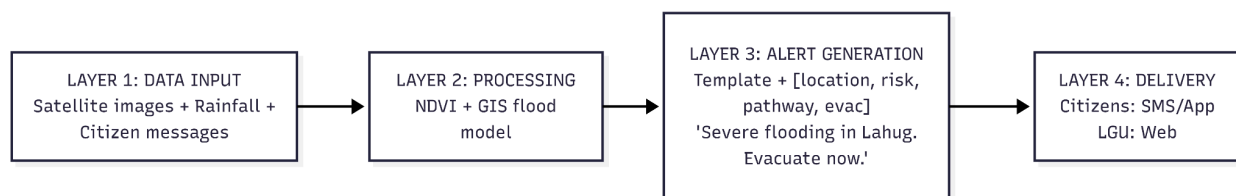


Figure 1. FloodGuard ASEAN system flow diagram.

V. CONCLUSION

This paper presents the NLP component of FloodGuard ASEAN, which integrates multilingual citizen reporting with satellite-based flood prediction. By combining keyword-based extraction with template-based text generation, the system provides an accessible, lightweight approach to processing citizen flood reports in real time. The use of keyword matching and template-based synthesis allows the system to operate without complex machine learning models, making it deployable on low-bandwidth networks and maintainable by local authorities. Future work will focus on expanding the keyword dictionary, improving accuracy on code-mixed text, and integrating real-time rainfall and satellite data to enhance flood pathway prediction.

5. REFERENCES

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6. APPENDIX

Appendix A: Sample Messages and Outputs

Sample 1 — High Urgency (Cebuano)

- Input: "Baha na sa Lahug tabang"
- Detected: Location = Lahug | Urgency = HIGH | Language = Cebuano
- Output: "FLOOD ALERT: Severe flooding is expected in Lahug within 6 hours. Floodwaters will move along Gorordo Avenue. Please evacuate to Cebu City Sports Center immediately."

Sample 2 — Medium Urgency (Tagalog)

- Input: "Mataas na ang tubig sa Guadalupe"
- Detected: Location = Guadalupe | Urgency = MEDIUM | Language = Tagalog
- Output: "FLOOD WARNING: Flooding is developing in Guadalupe. Water may rise within 4 hours. Move to the nearest evacuation center as a precaution."

Sample 3 — Low Urgency (English)

- Input: "Possible flooding in Talamban area"
- Detected: Location = Talamban | Urgency = LOW | Language = English
- Output: "FLOOD ADVISORY: Light flooding is possible in Talamban. Avoid low-lying areas and stay alert."

Appendix B: Evaluation Targets

- Location Extraction F1-Score: at least 0.75
- Urgency Classification F1-Score: at least 0.75
- Alert Clarity Rating (1 to 5): average of 4.0 or above
- System Usability Scale (SUS): 70 or above
- API response time: under 500 milliseconds

